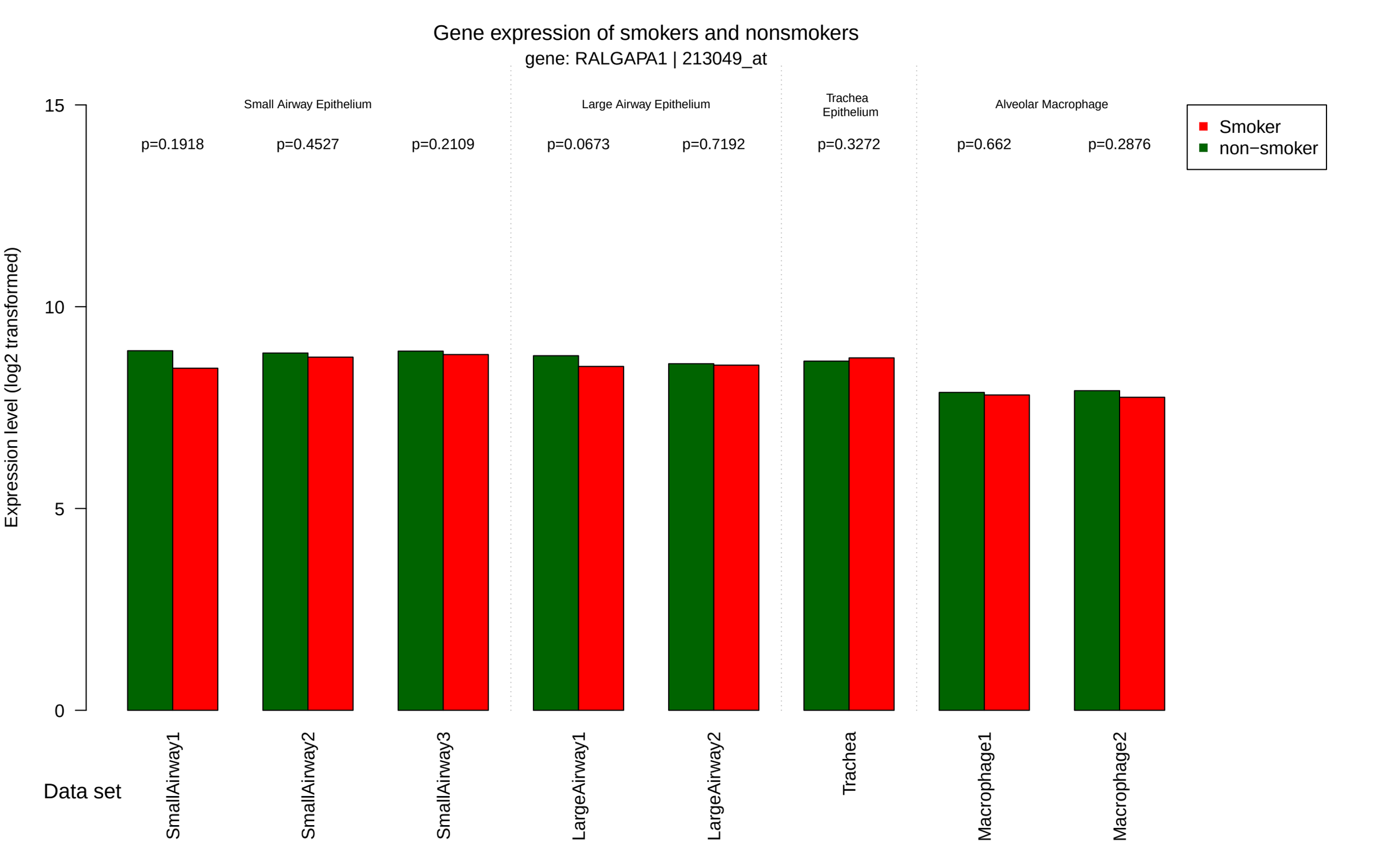
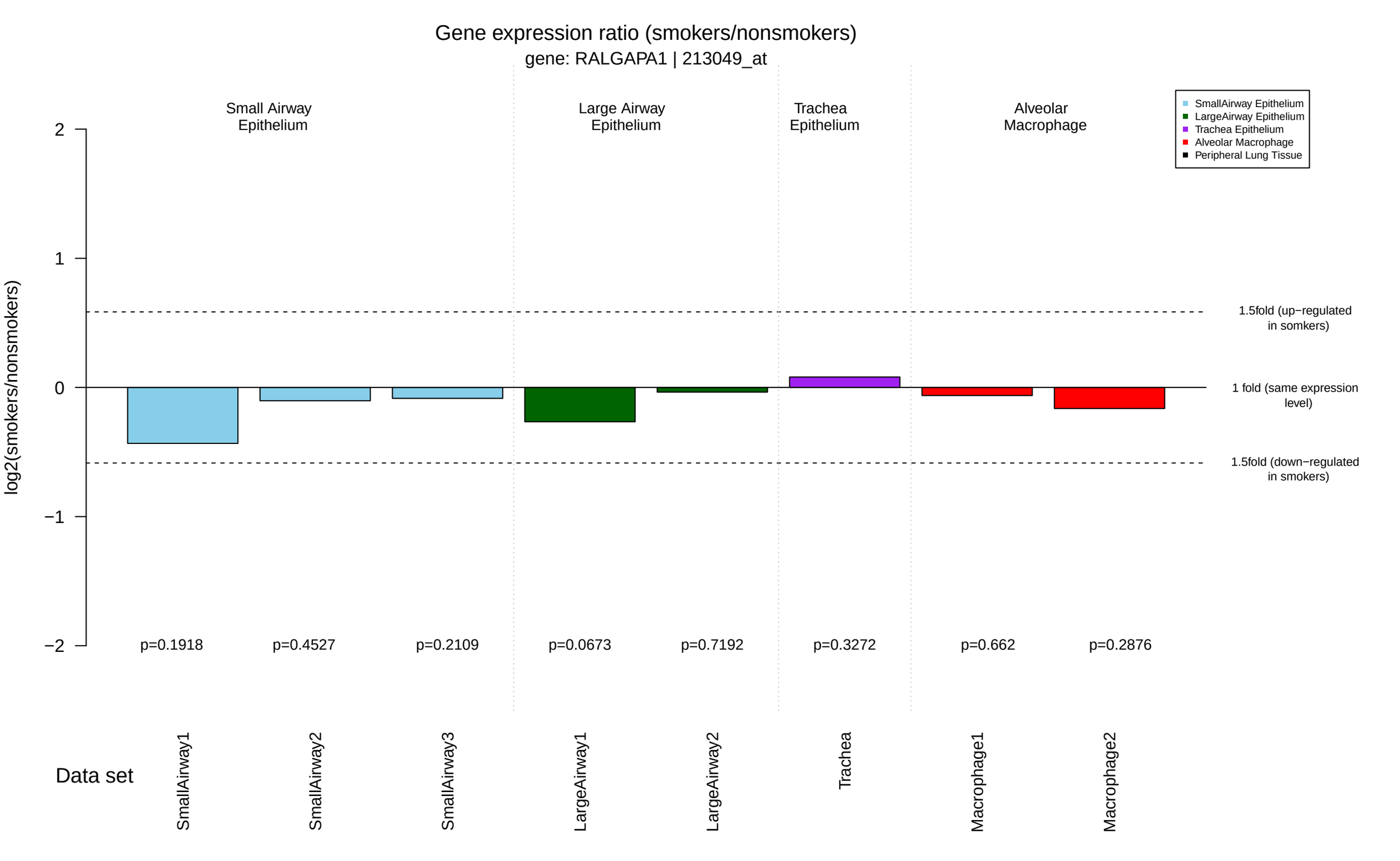
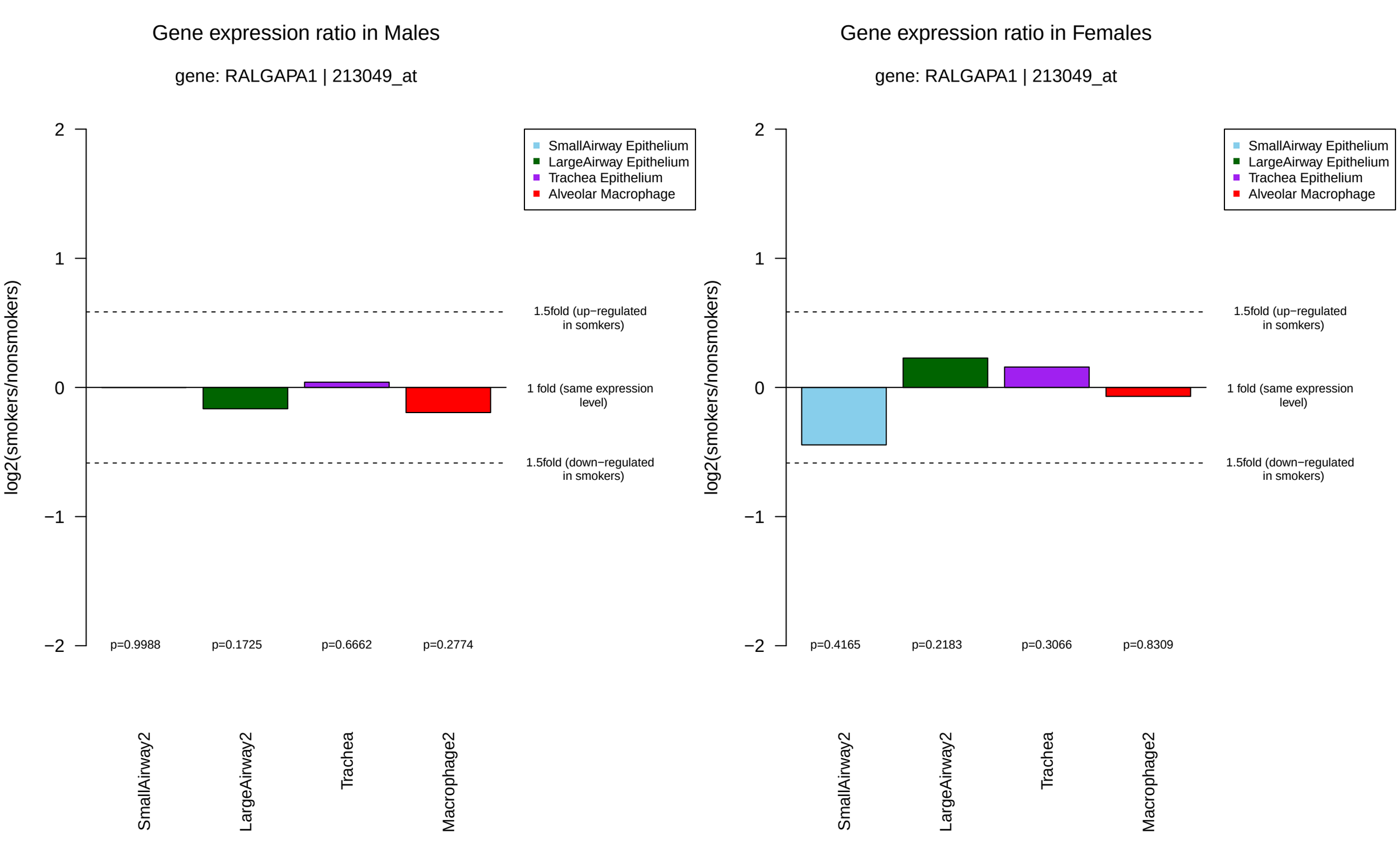
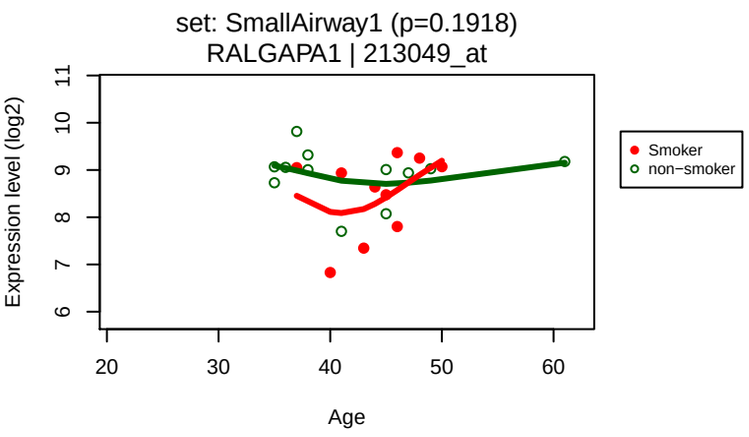
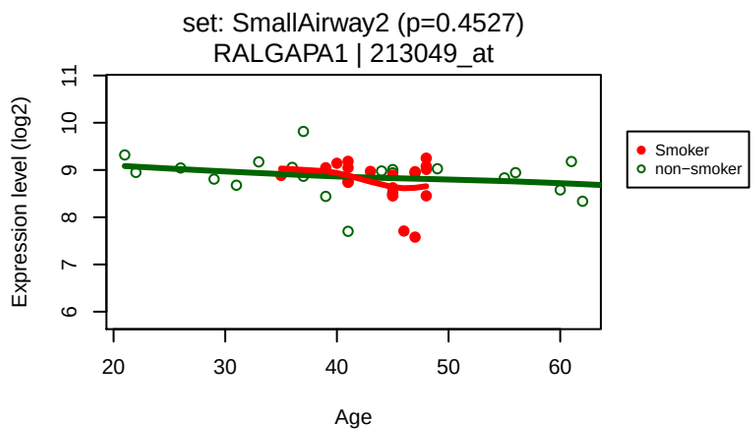
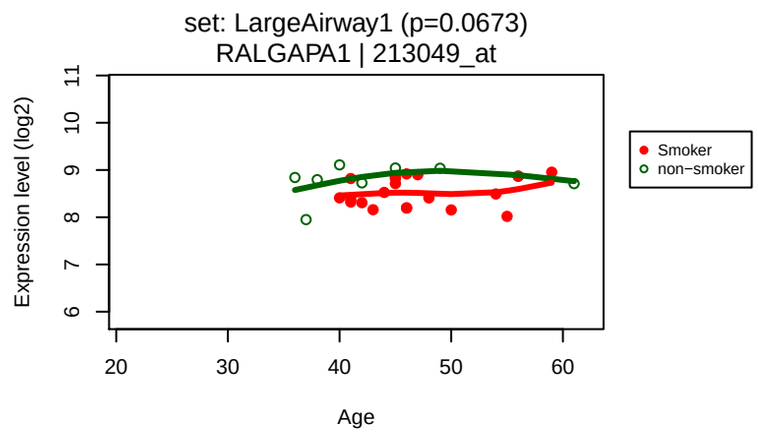

Gene expression of smokers and nonsmokers
gene: RALGAPA1 | 213049_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1918
p=0.4527
p=0.2109
p=0.0673
p=0.7192
p=0.3272
p=0.662
p=0.2876
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

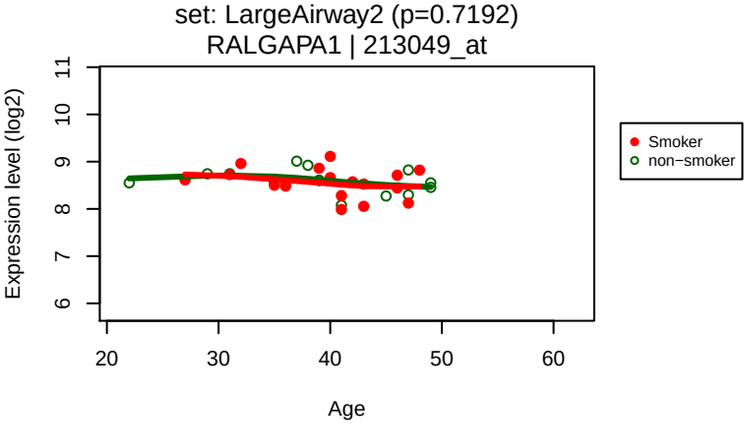

Gene expression ratio (smokers/nonsmokers)
gene: RALGAPA1 | 213049_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1918
p=0.4527
p=0.2109
p=0.0673
p=0.7192
p=0.3272
p=0.662
p=0.2876
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

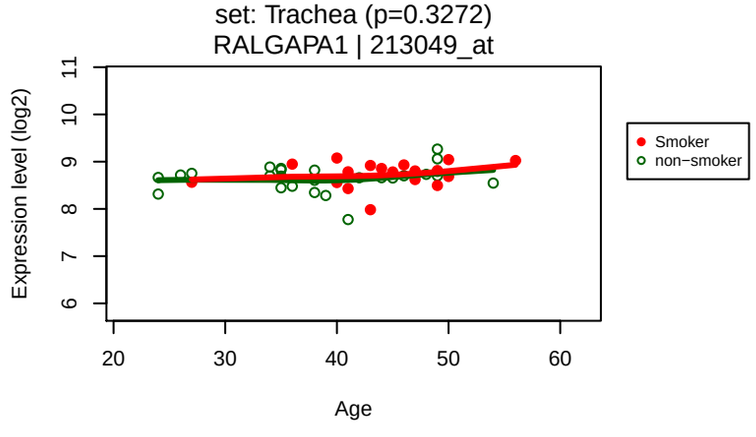

Gene expression ratio in Males
gene: RALGAPA1 | 213049_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9988
p=0.1725
p=0.6662
p=0.2774
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RALGAPA1 | 213049_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4165
p=0.2183
p=0.3066
p=0.8309
SmallAirway2
LargeAirway2
Trachea
Macrophage2

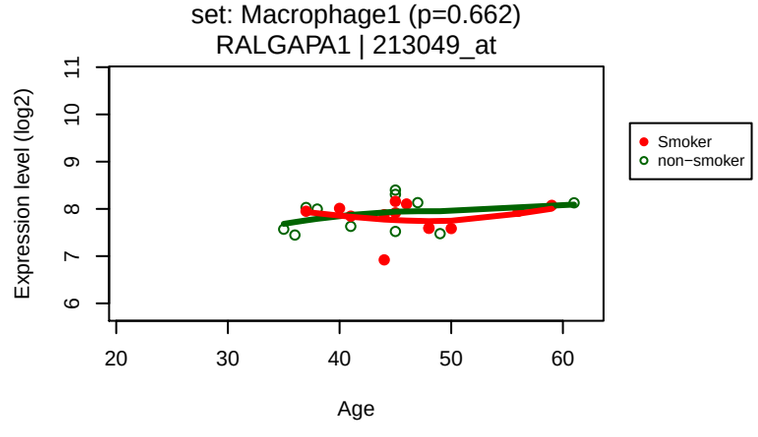

set: SmallAirway1 (p=0.1918)
RALGAPA1 | 213049_at
Expression level (log2)
Age
Smoker
non−smoker

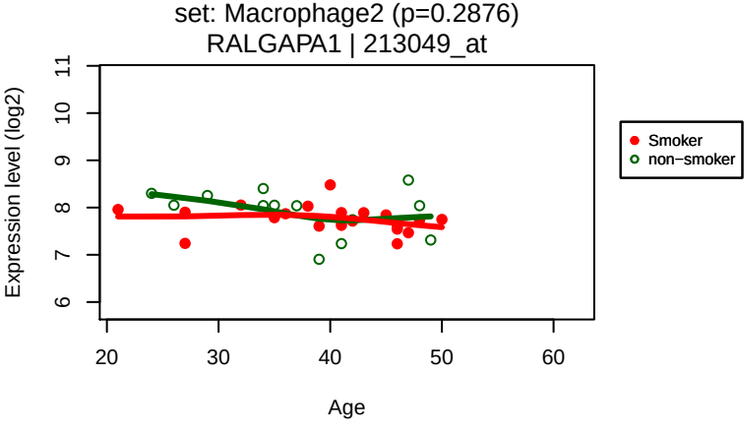

set: SmallAirway2 (p=0.4527)
RALGAPA1 | 213049_at
Expression level (log2)
Age
Smoker
non−smoker

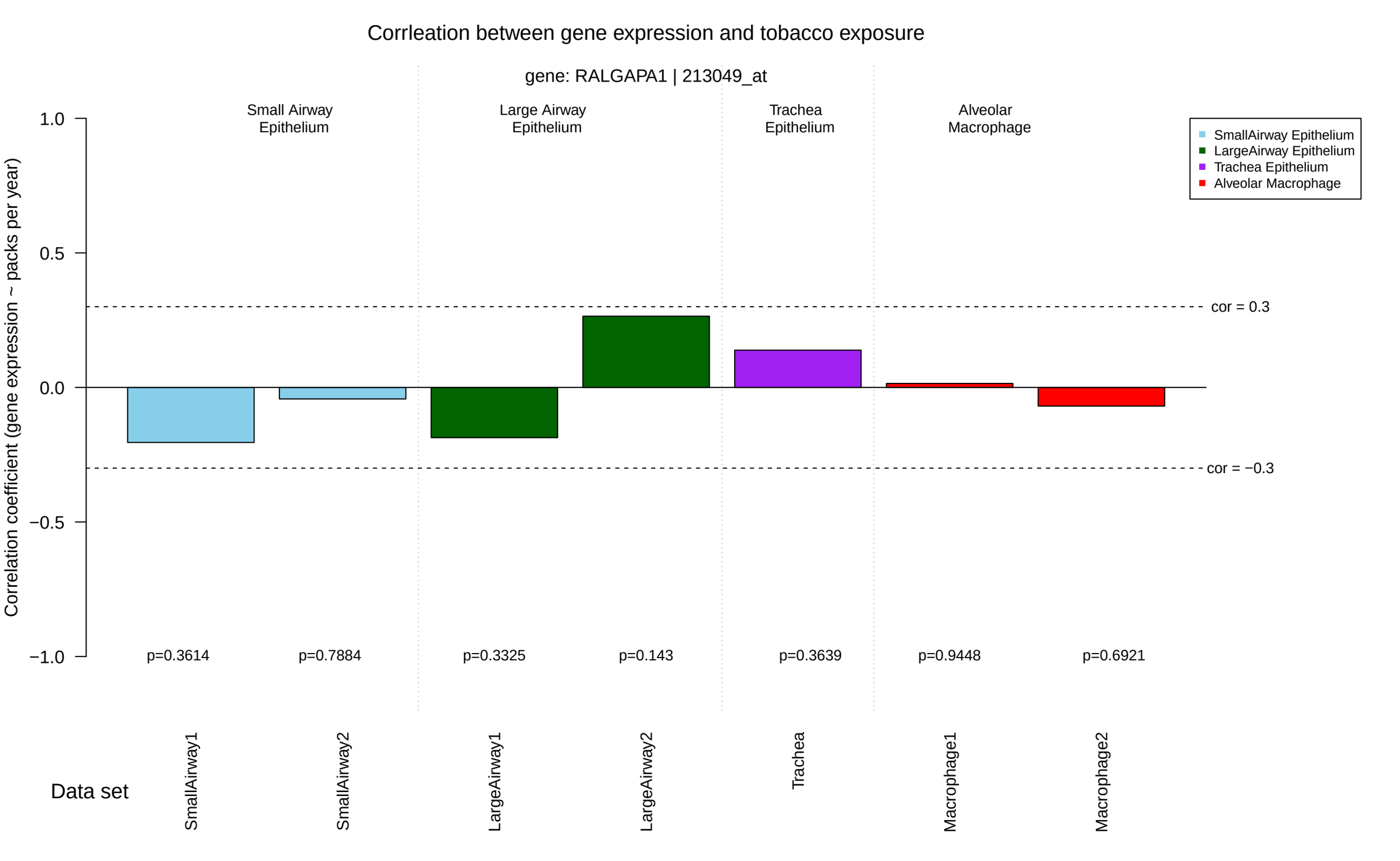

set: LargeAirway1 (p=0.0673)
RALGAPA1 | 213049_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7192)
RALGAPA1 | 213049_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3272)
RALGAPA1 | 213049_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.662)
RALGAPA1 | 213049_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2876)
RALGAPA1 | 213049_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RALGAPA1 | 213049_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.3614
p=0.7884
p=0.3325
p=0.143
p=0.3639
p=0.9448
p=0.6921
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2